SEMESTER S5
CLOUD COMPUTING

Course Code	**PEITT523**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT402 Computer Networks PCITT403 Operating Systems	**Course Type**	Theory

Course Objectives:

1. Gain a comprehensive understanding of cloud computing principles, including service and deployment models, and the key characteristics that define cloud computing.
2. Familiarize with the cloud computing reference architecture and understand the significance of virtualization in cloud computing.
3. Explore various cloud software environments and the architectural design of computing and storage clouds.
4. Learn about the security challenges in cloud computing, the role of cryptography, and various architectures to ensure secure data storage and processing in the cloud.
5. Gain proficiency in parallel and distributed programming models and develop practical skills in using major cloud platforms.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Cloud Computing – Introduction. Overview of Computing Paradigms-Grid Computing, Cluster Computing, Distributed Computing, Utility Computing. Cloud Service Models - Software as a Service, Platform as a Service, Infrastructure as a Service. Cloud Deployment Models - Private Cloud, Public Cloud, Community Cloud, Hybrid Cloud. Characteristics of Cloud Computing. Advantages and disadvantages of cloud computing. Cloud Computing Reference Architecture-Cloud Consumer, Cloud Provider, Cloud Auditor, Cloud Broker, Cloud Carrier, IBM Cloud Computing Reference Architecture. Cloud Computing Use Cases - Business Use Cases, Technical Use Cases	**9**
2	Virtualization- Introduction. Characteristics of Virtualization. Importance of Virtualization. Classification of Virtualization-Network Virtualization, Desktop Virtualization, Application Virtualization, Central Processing Unit Virtualization, Memory Virtualization, Input/Output Devices Virtualization. Virtualization to Private Cloud Services -Five High-Level Steps. Virtualization Architecture-Virtualization Platforms, Hypervisor. Tools and Mechanisms for Implementation of Virtualization. Physical and Virtual Clusters and Resource Management. Importance	**10**

	of Virtualization in Data Center Automation.	
3	Architectural Design of Compute and Storage Clouds-Data Center Interconnection Networks and Design, Architectural Design of Computing Clouds, Cloud Service Models and Platforms, Resource Management and Design Challenges. Fundamental Cloud Security-Threat agents, Cloud security threats, Operating system security, Infrastructure security- Network Level Security, Host Level Security, Application level security. Secure Distributed Data Storage in Cloud - Cryptography as a Primary Technique to Ensure Security, Gartner's Seven Cloud Computing Security Risks, Need for Data Security. Cloud Computing—Attacks. Cloud Data Security Requirements. Cloud Security—Controls. Cloud Security Architecture. Cloud Data Security.	**10**
4	Parallel and Distributed Programming Models. MapReduce- MapReduce Programming Model, Logical Data Flow and Control Flow, Case Study: MapReduce Program to Find Inverted Index of Documents. Twister and Iterative MapReduce, Hadoop Library, Apache Spark. Cloud Software Environments-Eucalyptus, OpenNebula, OpenStack, Aneka, CloudSim. Popular Cloud Databases – MongoDB, HBase. Popular Cloud Platforms. Amazon Web Services- Amazon Web Services (AWS)- Elastic Compute Cloud (EC2), Simple Storage Service (Amazon S3), Elastic Block Store (EBS), Amazon CDN services. Google Cloud Platform (GCP)- Compute Engine (GCE), Google App Engine (GAE), Cloud Storage, Gmail, Google Drive. Microsoft Azure- Azure Virtual Machine, Hyper-V, Azure Storage services.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe the fundamental concepts of cloud computing, including service and deployment models, and the cloud computing reference architecture.	**K2**
CO2	Explain the principles of virtualization and its role in enabling cloud services, including different types of virtualization and their applications.	**K2**
CO3	Identify the architectural components of compute and storage clouds and discuss basic security challenges in cloud computing.	**K2**
CO4	Implement parallel and distributed programming models using MapReduce and utilize cloud databases like MongoDB and HBase.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	2	1	-	-	-	-	-	-	2
CO2	3	2	-	2	1	-	-	-	-	-	-	2
CO3	2	1	-	2	2	-	-	-	-	-	-	3
CO4	3	3	2	3	2	-	-	-	-	-	-	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Cloud Computing & Big Data From the basics to practical use cases	Sudheep E. M., Sarith D. M., Lija M., Tanmay K. P. & Shubham A.	Cengage Learning	1st Edition, 2024
2	Cloud Computing Concepts, Technology & Architecture	Thomas, E., Zaigham M., & Ricardo P.	Prentice Hall	1st Edition, 2013
3	Mastering cloud computing: foundations and	Buyya, R., Vecchiola, C., & Selvi, S. T.	Morgan Kaufmann	1st Edition, 2017

	applications programming			
4	Cloud computing	Bhowmik, S.	Cambridge University Press	1st Edition, 2017

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Cloud computing: theory and practice	Marinescu, D. C.	Morgan Kaufmann	2nd Edition, 2017
2	Cloud computing: Principles and paradigms	Buyya, R., Broberg, J., & Goscinski, A. M.	John Wiley & Sons.	1st Edition, 2011

Video Links (NPTEL, SWAYAM…)	
No.	**Link ID**
1	https://youtu.be/NzZXz3fJf6o
2	https://youtu.be/fZ3D6HQrWzs
3	https://youtu.be/LcAPj95KeSA
4	https://youtu.be/TOOSVsxEIpo